More Trade, More Dispute? Measuring Alliance Effect on WTO Conflict Dataset Codebook

Peng-Ting Kuo

March 22, 2026

Abstract

This codebook documents all datasets used in the paper *More Trade, More Dispute: Measuring Alliance Effect on WTO Conflict* (March 2026). The core country-year panel covers 196 polities over 1995–2024 (5,880 observations, 77 variables), integrating seven data sources: WDI, WGI, UN GA Ideal Points (Bailey, Strezhnev, and Voeten 2017), V-Dem v15, COW NMC 6.0, ATOP 5.1, and WTO dispute records. The dyadic dataset (`bilateral_trade_wto.csv`, 545k rows) merges BACI trade, ATOP alliances, and DESTA trade agreement depth. Case-level severity scoring and HS classification are available for DS1–DS626 via the RAG pipeline. Four ERGM-ready analysis datasets are documented in Section 6. Systematic missing data by source is in Section 2.

Contents

1 Panel Dataset **3**
- 1.1 Panel Structure . . . 3
- 1.2 A. Identifiers . . . 3
- 1.3 B. WTO Membership & Participation . . . 3
- 1.4 C. EU / Eurozone Membership . . . 4
- 1.5 D. UN Voting Ideal Points . . . 4
  - 1.5.1 D.1 Ideal Point Estimates . . . 5
  - 1.5.2 D.2 Vote Counts and Posterior Quantiles . . . 5
- 1.6 E. V-Dem Democracy Indicators . . . 5
- 1.7 F. World Development Indicators (WDI) – Economic . . . 6
- 1.8 G. Worldwide Governance Indicators (WGI) . . . 7
- 1.9 H. Geographic & Classification Variables . . . 8
- 1.10 I. Derived Variables . . . 9
- 1.11 J. COW National Material Capabilities (NMC) . . . 9

2 Systematic Missing Data **10**
- 2.1 UN Voting Ideal Points . . . 10
- 2.2 V-Dem Democracy Indices . . . 10
- 2.3 COW National Material Capabilities . . . 10
- 2.4 World Development Indicators (WDI) . . . 11
- 2.5 WGI Governance Indicators . . . 11
- 2.6 ERGM Datasets . . . 12

3 Data Processing & Construction Notes **12**
- 3.1 Unit Standardization . . . 12
- 3.2 WGI Interpolation for Missing Years . . . 12
- 3.3 WTO Participation Variables . . . 13
- 3.4 EU Representation in Trade Data . . . 13
- 3.5 EU Disaggregation Rules for ERGM Datasets . . . 13

4 WTO Case Data **14**
- 4.1 Case Dataset . . . 14
- 4.2 RAG Pipeline Outputs . . . 15
  - 4.2.1 Industry Extraction . . . 15
  - 4.2.2 HS Classification . . . 16
  - 4.2.3 HS Expansion . . . 16
  - 4.2.4 Severity Score . . . 16
- 4.3 Network Analysis Results . . . 17

5 Dyadic Datasets **18**
- 5.1 WTO Dispute Dyads . . . 18
- 5.2 Bilateral Trade (WTO-filtered) . . . 18
  - 5.2.1 Trade Variables (from CEPII BACI HS92) . . . 18
  - 5.2.2 ATOP Alliance Variables . . . 19
  - 5.2.3 DESTA Trade Agreement Variables . . . 19
- 5.3 Bilateral Trade by Section (WTO-filtered) . . . 19
- 5.4 DESTA Trade Agreement Panel . . . 20

6 ERGM Analysis Datasets **20**
6.1 `wto_cases_enriched.csv` (644 rows, 44 cols) 21
6.2 `wto_dyadic_enriched.csv` (8,373 rows, 64 cols) 21
6.3 `ergm_dyad_year_eun.csv` (545,344 rows, 190 cols) 21
6.4 `ergm_dyad_year_eu_disagg.csv` (537,833 rows, 190 cols) 22

Panel Dataset

`country_meta_1995_2024.csv`

Panel Structure

Attribute	Value
Unit of analysis	Country-year
Observations	5,880
Countries / Polities	196
Year range	1995–2024 (30 years)
Primary key	`(country, year)`
Total variables	77

Table 1: Panel dataset overview.

The dataset integrates six major data sources into a balanced country-year panel covering all WTO-relevant polities (members, observers, and non-members that appear in WTO dispute records), plus the EU as a collective actor.

A. Identifiers

Variable	Definition	Unit / Range	Source
`country`	Country / polity name	String	User-defined
`iso3c`	ISO 3166-1 alpha-3 code	3-letter	WDI / User
`ccode`	COW numeric code; -1 if no match	Integer	COW
`stateabb`	COW 3-letter abbreviation	String	COW
`un_country_name`	Official UN name	String	UN / WDI
`year`	Calendar year	1995–2024	—

Table 2: Identifier variables.

B. WTO Membership & Participation

Source: WTO official membership records and `wto_cases_v2.csv` (644 cases DS1–DS644, 1995–2024).

Variable	Definition	Range	Notes
`wto_member`	Member in that year	Binary (0/1)	112 (1995) to 166 (2024)
`wto_accession_year`	Year of WTO accession	1995–2024	`NaN` for non-members
`complainant`	Filed ≥1 dispute that year	Binary (0/1)	—
`respondent`	Faced ≥1 dispute that year	Binary (0/1)	—
`third_party`	Joined as third party that year	Binary (0/1)	—
`wto_participant`	Any dispute activity	Binary (0/1)	OR of above three
`cum_complainant`	Cumulative **distinct cases** filed as complainant (1995–t)	0–124	US: 124; EU: 113
`cum_respondent`	Cumulative **distinct cases** faced as respondent (1995–t)	0–158	US: 158; EU: 96
`cum_third_party`	Cumulative **distinct cases** joined as third party (1995–t)	0–234	JPN: 234; EU: 219

Table 3: WTO membership and dispute participation variables.

C. EU / Eurozone Membership

Variable	Definition	Values	Notes
`eu_member`	EU member in that year	Binary (0/1)	28 unique countries ever
`euro_join_year`	Year of Euro adoption	1999–2023	`NaN`: 89.8% of obs
`euro_member`	Euro user in that year	Binary (0/1)	Derived from above

Table 4: EU and Eurozone membership variables. Source: EU official records; ECB Eurozone records.

D. UN Voting Ideal Points

Primary citation: Bailey, Michael A., Anton Strezhnev, and Erik Voeten. 2017. "Estimating Dynamic State Preferences from United Nations Voting Data." *Journal of Conflict Resolution* 61 (2): 430–456.

Data version: Fjelstul, Joshua, Simon Hug, and Christopher Kilby. 2025. "Decision-making in the United Nations General Assembly: A comprehensive database of resolution-related decisions." *The Review of International Organizations.*
Dataset: https://dataverse.harvard.edu

Ideal points are estimated using a Bayesian IRT model on UNGA voting records. Higher values indicate alignment with the United States; lower values indicate alignment with Russia/China.

D.1 Ideal Point Estimates

Variable	Definition	Range
`idealpointfp`	Ideal point – final-passage votes only (preferred for most analyses)	−2.10 to +3.03
`idealpointall`	Ideal point – all votes incl. paragraphs and amendments; may over-weight specific issues (e.g., Gaza)	−2.36 to +3.06
`idealpointlegacy`	Legacy session-based estimation; $r = 0.988$ with `idealpointall`	−2.17 to +3.22

Table 5: UN Voting ideal point estimates.

D.2 Vote Counts and Posterior Quantiles

Variable	Definition	Range
`nvotesfp`	Number of final-passage votes used for FP estimate	1–109
`nvotesall`	Number of all votes used for ALL estimate	1–232
`nvotesLegacy`	Number of votes for legacy estimate	1–96
`q5fp`–`q100fp`	5th–100th percentile of MCMC posterior (FP model)	≈−2.4 to +3.7
`q5all`–`q100all`	5th–100th percentile of MCMC posterior (All model)	≈−2.6 to +3.5

Table 6: Vote counts and posterior uncertainty quantiles. `qofp` / `qoall` = posterior mean (equivalent to point estimates above).

E. V-Dem Democracy Indicators

Citation: Coppedge, Michael, et al. 2025. *V-Dem Dataset v15.*
Pemstein, Daniel, et al. 2025. *The V-Dem Measurement Model: Latent Variable Analysis for Cross-National and Cross-Temporal Expert-Coded Data.* Technical report V-Dem Working Paper No. 21. University of Gothenburg: Varieties of Democracy Institute.
Dataset: https://www.v-dem.net/data

Variable	Definition	Range	Codebook ref.
`v2x_polyarchy`	Electoral Democracy Index: freedom of expression, clean elections, suffrage, elected officials	$[0, 1]$	v15, p. 37
`v2x_libdem`	Liberal Democracy Index: electoral democracy plus minority rights protection	$[0, 1]$	v15, p. 38
`v2x_regime`	Regime type: 0 = Closed Autocracy; 1 = Electoral Autocracy; 2 = Electoral Democracy; 3 = Liberal Democracy	0–3	v15, p. 291
`v2eltype_0`	Legislative election held	0/1/`NaN`	Election years only
`v2eltype_1`	Presidential election held	0/1/`NaN`	Election years only
`v2eltype_6`	Constituent assembly election	0/1/`NaN`	Election years only
`v2eltype_7`	Other executive election	0/1/`NaN`	Election years only
`is_leg_elec`	Any legislative election (derived)	Binary (0/1)	—
`is_pres_elec`	Any presidential election (derived)	Binary (0/1)	—
`election_binary`	Any election, legislative or presidential	Binary (0/1)	—

Table 7: V-Dem indices and election variables. `NaN` in `v2eltype_*` denotes a non-election year, not missing data.

F. World Development Indicators (WDI) – Economic

Source: World Bank, World Development Indicators (WDI), accessed 2025. https://databank.worldbank.org/source/world-development-indicators

Taiwan supplements: GDP/GDPPC from DGBAS (Executive Yuan, Taiwan); FDI from UNCTAD Global Investment Data.

Unit standardization: Monetary values ÷ 1,000,000 → **Million USD**. WDI percentage values ÷ 100 → **Ratio (0–1)**.

Variable	Definition (WDI indicator code)	Unit	Range
`gdp`	Real GDP, constant 2015 USD (`NY.GDP.MKTP.KD`)	M USD	23.6 – 22,568,000
`gdppc`	Real GDP per capita, constant 2015 USD (`NY.GDP.PCAP.KD`)	USD	211 – 247,170
`pop`	Total population (`SP.POP.TOTL`)	Count	9,280 – 1.45B
`gdp_growth_rate`	Annual GDP growth rate (`NY.GDP.MKTP.KD.ZG`)	Ratio	−0.50 – +1.50
`trade`	(Exports + Imports) / GDP (`NE.TRD.GNFS.ZS`)	Ratio	0.0002 – 4.37
`fdi`	Net FDI inflows / GDP (`BX.KLT.DINV.WD.GD.ZS`)	Ratio	−1303 – +1283
`fdi_inflow_usd`	Net FDI inflows, absolute (`BX.KLT.DINV.CD.WD`)	M USD	−343k – +734k
`exp_share`	Exports of goods and services / GDP (`NE.EXP.GNFS.ZS`)	Ratio	0.000054 – 2.29
`imp_share`	Imports of goods and services / GDP (`NE.IMP.GNFS.ZS`)	Ratio	0.000156 – 2.08
`unemployment_rate`	Unemployment rate, national estimate (`SL.UEM.TOTL.NE.ZS`)	Ratio	0.00039 – 0.39

Table 8: WDI economic variables. `fdi` ratio can exceed ±100% for small open economies. Full FDI range: $-343,400$ to $+733,830$ M USD.

G. Worldwide Governance Indicators (WGI)

Citation: Kaufmann, Daniel, and Aart Kraay. 2025. "The Worldwide Governance Indicators : Methodology and 2024 Update." *SSRN Electronic Journal* https://info.worldbank.org/governance/wgi/

All six indicators are on a standard normal scale (−2.5 to +2.5); higher = better governance.

Variable	WGI Code	Definition	Range
`voice`	VA.EST	Voice and Accountability: citizen participation in selecting government; freedom of expression, association, and press	−2.31 to +1.80
`stability`	PV.EST	Political Stability and Absence of Violence/Terrorism	−3.18 to +1.76
`efficiency`	GE.EST	Government Effectiveness: quality of public services, civil service, and policy formulation	−2.44 to +2.47
`reg_quality`	RQ.EST	Regulatory Quality: formulating policies that promote private sector development	−2.53 to +2.31
`law`	RL.EST	Rule of Law: contract enforcement, property rights, courts, police	−2.33 to +2.12
`corruption`	CC.EST	Control of Corruption: extent to which public power is exercised for private gain	−1.97 to +2.46

Table 9: WGI governance indicators (higher = better). Pre-2002 gaps interpolated: 1995 backfilled from 1996; 1997, 1999, 2001 linearly interpolated.

H. Geographic & Classification Variables

Variable	Definition	Values / Unit
`region`	World Bank geographic region	7 categories: East Asia & Pacific; Europe & Central Asia; Latin America & Caribbean; Middle East & North Africa; North America; South Asia; Sub-Saharan Africa. `NaN` for EU.
`longitude`	Geographic centroid longitude	Decimal degrees; −175.2 to +179.1
`latitude`	Geographic centroid latitude	Decimal degrees; −41.3 to +121.0
`income`	World Bank income group	Low / Lower-middle / Upper-middle / High / Not classified. `NaN` for EU.

Table 10: Geographic and income classification. Source: WDI Extra fields; user-defined for Taiwan and EU.

I. Derived Variables

Variable	Definition	Formula	Range
`log_gdppc`	Natural log of real GDP per capita	ln(`gdppc`)	5.35 – 12.42
`log_pop`	Natural log of total population	ln(`pop`)	9.14 – 21.10

Table 11: Derived variables constructed within the project; no external source.

J. COW National Material Capabilities (NMC)

Citation: Singer, J. David. 1988. "Reconstructing the correlates of war dataset on material capabilities of states, 1816–1985." _Eprint: https://doi.org/10.1080/03050628808434695, *International Interactions* 14 (2): 115–132.
Singer, J. David, Stuart A. Bremer, and John Stuckey. 2012. "Capability distribution, uncertainty, and major power war, 1820–1965 1." In *Advancing Peace Research,* 161–174. Routledge.

Dataset: NMC 6.0 (1816–2016). https://correlatesofwar.org/data-sets/

Critical coverage note: NMC 6.0 ends in **2016**. All 196 countries have `NaN` for all NMC variables for 2017–2024 (1,568 missing observations total). Thus the index is only use for robustness check purpose.

Var.	Definition	Unit	Range / Notes
`cinc`	Composite Index of National Capability: unweighted average of six capability shares. Codebook p. 4: "CINC is the average of a state's share of the system total for each of the six capability components."	Proportion (0–1)	2.4×10^{-7} – 0.231
`milex`	Military expenditures	Thousands USD (post-1914); thousands GBP (pre-1914)	-9 to $693,600,000$
`milper`	Military personnel	Thousands of persons	-9 to $2,930$
`pec`	Primary energy consumption	Thousands coal-ton equivalents	-9 to $5,900,300$
`tpop`	Total population (COW measure)	Thousands of persons	9 – $1,403,500$
`upop`	Urban population (cities > 100k; > 300k after 2002)	Thousands of persons	0 – $612,930$

Table 12: COW NMC variables. Raw sentinel -9 = missing; recode to `NaN` before modelling. All values are `NaN` for 2017–2024.

Systematic Missing Data

UN Voting Ideal Points

Polity (all 30 years)	Reason for exclusion
Taiwan	Not a UN member (PRC replaced ROC in 1971); no UNGA voting record
Hong Kong	Not a sovereign state; votes subsumed under China (PRC)
Macao, China	Not a sovereign state; votes subsumed under China (PRC)
EU	Not a UN member state; EU participates as observer only

Table 13: Polities entirely missing UN Voting ideal points across all years.

Partially missing (some years, due to non-membership or absent data): Afghanistan, Bosnia & Herzegovina, Dominican Republic, Gambia, Guinea-Bissau, Iraq, Kiribati, Kyrgyzstan, Liberia, Mauritania, Montenegro, Nauru, Niger, Palau, Sao Tome and Principe, Seychelles, Somalia, South Sudan, Switzerland (observer until 2002), Tajikistan, Timor-Leste, Tonga, Tuvalu, Uzbekistan, Vanuatu, Venezuela.

V-Dem Democracy Indices

Polities entirely absent from V-Dem v15 (all 30 years): EU, Hong Kong, Macao, Andorra, Monaco, San Marino, Liechtenstein, Antigua and Barbuda, Bahamas, Belize, Brunei Darussalam, Dominica, Grenada, Kiribati, Marshall Islands, Micronesia (Fed. States), Montenegro, Nauru, Palau, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Samoa, South Sudan, Tonga, Tuvalu.

COW National Material Capabilities

- **2017–2024 (all 196 countries):** NMC 6.0 ends in 2016.
- **Additional historical gaps:** EU, Hong Kong, Macao, micro-states (Andorra, Monaco, San Marino, Liechtenstein), South Sudan (pre-2011), Montenegro (pre-2006).

World Development Indicators (WDI)

Polity	Reason
North Korea	No WDI reporting; data isolation and international sanctions
Syria	Post-2011 conflict severely disrupted statistical capacity
Somalia	Prolonged state fragility; limited statistical infrastructure
South Sudan	Newly independent (2011); early years lack data
Lebanon	Severe economic crisis (2019–) disrupted reporting
EU	WDI covers member states; not the supranational entity
Liechtenstein, Monaco, San Marino	Small states not covered by WDI
Taiwan	Not a WDI member; supplemented from DGBAS

Table 14: Countries with substantial WDI gaps. `unemployment_rate` missing for 42.8% of obs (ILO non-reporting by developing countries). `trade`, `exp_share`, `imp_share` missing for 18.1%.

WGI Governance Indicators

Missing ≈5–9% of observations. Systematic gaps:

- **North Korea:** Not covered.
- **Small island states** (Kiribati, Nauru, Tuvalu, Palau, etc.): Partial coverage only.
- **EU:** Supranational entity; not applicable.
- **Pre-2002 years:** Interpolated (see Section 3).

ERGM Datasets

Variable Group	Affected Obs.	Reason
Trade lags `*_t1/t2/t3`	1995 rows: 100% NaN; 1996: t-2/t-3 imputed; 1997: t-3 imputed	No pre-1995 BACI data; backward imputation applied
UN ideal points	Taiwan (TAW), HK, Macao, EUN: all 30 years NaN	Not UN members
V-Dem indices	EU, HK, Macao, micro-states: all years NaN	Outside V-Dem coverage
CINC	All 196 countries, 2017–2024	NMC v6.0 ends 2016
WGI	North Korea, very small island states, EUN (before aggregation)	Not covered by WGI
Severity scores	DS627–DS644 (18 cases): NaN	No PDF documents; RAG covers DS1–DS626 only

Table 15: Systematic missing data in ERGM panel datasets. EUN node attributes (gdp, pop, V-Dem, WGI) are aggregated from individual EU member states in the ERGM pipeline; raw `country_meta` has NaN for EUN on these fields.

Data Processing & Construction Notes

Unit Standardization

Transformation	Details
GDP / FDI monetary values	Divided by 1,000,000 → Million USD
WDI percentage variables	Divided by 100 → Ratio (0–1); applies to `trade`, `fdi`, `exp_share`, `imp_share`, `unemployment_rate`, `gdp_growth_rate`
COW NMC sentinel values	Raw −9 retained as-is in the dataset; must be recoded to `NaN` before modelling

Table 16: Unit standardization applied during data construction.

WGI Interpolation for Missing Years

WGI surveys were not published annually before 2002:

- **1995:** Backward fill using 1996 values.
- **1997:** Linear interpolation of 1996 and 1998 values.
- **1999:** Linear interpolation of 1998 and 2000 values.
- **2001:** Linear interpolation of 2000 and 2002 values.

WTO Participation Variables

`complainant`, `respondent`, `third_party`, `wto_participant` are binary indicators from `wto_cases_v2.csv` (644 cases): 1 if the country participated in ≥1 dispute of that type in that year.

`cum_complainant`, `cum_respondent`, `cum_third_party` count the **cumulative number of distinct cases** from 1995 through year t inclusive. If a country filed 3 cases in one year the cumulative count rises by 3 that year (not by 1). Computed via `nunique(case_id)` per (country, year, role) from `wto_dyadic_v2.csv` before cumsum. Updated March 2026.

EUN participation data: EUN's `complainant`, `respondent`, `cum_*` etc. are sourced from EUN's own rows in `country_meta` (collective EU filings under "European Communities" / "European Union"), *not* aggregated from individual member states.

The ERGM pipeline adds **annual** (non-cumulative) counts: `n_complainant_t`, `n_respondent_t`, `n_tp_t` = number of distinct cases each country participated in during that calendar year, merged from `wto_dyadic_v2.csv` consultation years.

EU Representation in Trade Data

Dual EU representation in `bilateral_trade_wto.csv`: Individual EU member states retain their own trade flows (including intra-EU), while a separate `EUN` aggregate captures the EU as a single external actor (intra-EU trade excluded). BACI assigns EU-external trade to `EUN`; intra-EU trade appears under individual country codes.

ATOP forward-fill: ATOP v5.1 ends in 2018. Alliance values from 2018 are forward-filled to 2019–2024 for all dyads.

DESTA EUN data artifact: DESTA records EU trade agreements through individual member states (e.g., "EC–Korea FTA" appears as DEU→KOR, FRA→KOR, etc.). Consequently, all 3,754 EUN rows in `bilateral_trade_wto.csv` have `label=0`. The ERGM pipeline corrects this post-hoc in `fix_eun_desta()`: for each (EUN, partner, year), if any EU member has `label=1` with that partner, EUN inherits the highest-depth agreement. `bilateral_trade_wto.csv` is left unchanged.

EU Disaggregation Rules for ERGM Datasets

`ergm_dyad_year_eu_disagg.csv` replaces EUN rows with individual EU member state rows. Three categories exist in `wto_dyadic_v2.csv`:

Category	Cases	EUN rows	Treatment
EUN-only (no individual members listed)	395	Yes	Expand: propagate `has_dispute=1` to all individual EU member rows in that year; drop EUN rows
Mixed (EUN + individual members, e.g. DS316 Airbus)	37	Yes	Keep individual member rows as-is; drop EUN rows
Individual-EU-only (no EUN, e.g. DS37 USA vs PRT)	30	No	Unchanged

Table 17: EU disaggregation rules. DS316 (Airbus) is an example of a mixed case where both EUN and specific members (France, Germany, Spain, UK) are listed as respondents. The `eu_disagg` version retains 537,833 rows after removing 7,511 EUN rows.

WTO Case Data

Case Dataset

`wto_cases_v2.csv`

Attribute	Value
Rows	644 (DS1–DS644)
Columns	29
Year range	1995–2026
Script	`scripts/scrape_wto_cases.py` (March 2026)

Table 18: `wto_cases_v2.csv` overview. Country names harmonized: European Communities → European Union (230 cases); Turkey → Türkiye (119 cases).

Variable	Definition
`case`	Case number string (`DS1`–`DS644`)
`title`	Full case title ("Respondent — Product" format)
`short_title`	Short case name
`complainant`	Complainant country/entity
`respondent`	Respondent country/entity
`third_parties`	Semicolon-separated list of third parties
`product`	Product/subject extracted from title (ground truth for HS classification)
`dispute_stage`	6 categories: Consultation, Mutually Agreed Solution, Panel, Appellate Body, Implementation, Retaliation
`current_status`	Free-text current status
`summary`	Case summary
`agreements_cited_*`	WTO agreements cited at consultation / panel stage
13 procedural date columns	`consultations_requested`, `panel_requested`, `panel_established`, `panel_composed`, `panel_report_circulated`, `appellate_body_report_circulated`, `mutually_agreed_solution_notified`, Art. 21.5 / 22.6 / 25 dates, `panel_lapsed`, etc.

Table 19: Key variables in `wto_cases_v2.csv`

Dispute stage distribution (644 cases):

Dispute Stage	Cases	Share
Consultation	198	30.7%
Panel	75	11.6%
Appellate Body	74	11.5%
Mutually Agreed Solution	119	18.5%
Implementation	153	23.8%
Retaliation	25	3.9%

Table 20: Dispute stage distribution (DS1–DS644). Counts are approximate due to ongoing scraping of recent cases.

RAG Pipeline Outputs

(`Data/Output/`) All four files cover DS1–DS626 (626 cases with PDF documents). Cases DS627–DS644 have no RAG outputs.

Industry Extraction

`industry_extraction.csv` (626 rows, 11 cols)

Variable	Definition
`case_id`	Integer case number
`product_descriptions`	RAG-extracted product descriptions (pipe-separated)
`explicit_hs_codes`	Explicitly cited HS codes found in documents
`is_systemic`	Dispute concerns a systemic policy measure (0/1)
`is_services`	Dispute concerns services / GATS (0/1)
`confidence`	Extraction confidence level
`notes`, `n_parents_retrieved`	Retrieval metadata

Table 21: Stage 1 RAG output: product/industry extraction.

HS Classification

`case_hs_sections.csv` (626 rows, 12 cols)

Variable	Definition
`case_id`	Integer case number
`title_hs_sections`	HS sections from `product` column (ground truth; pipe-sep. 1–21)
`title_case_type`	Case type from title: `product` or `policy`
`hs_sections`	HS sections from RAG+LLM classification (pipe-sep. 1–21)
`case_type`	LLM-assigned type: `product` or `policy`
`product_descriptions`	Descriptions for product cases
`policy`	Policy description for policy cases
`extraction_method`	`explicit_hs`, `llm_classification`, or `title_fallback`
`confidence`, `reasoning`	LLM quality metadata

Table 22: Stage 2 RAG output: HS section classification. Horizontal policy cases span all 21 sections; `hs_sections` has no missing values.

HS Expansion

`case_section_expanded.csv` (3,003 rows, 5 cols) One row per case-section pair (primary key: `case_id` × `hs_section`). Columns: `case_id`, `hs_section` (1–21), `case_type`, `extraction_method`, `confidence`. Used by the ERGM pipeline for section-level trade lookups and HS section dummies.

Severity Score

`severity_scores_raw.csv` (626 rows, 10 cols)

Variable	Range	Definition
`case_id`	1–626	Integer case number
`complainant`	—	Complainant country name
`rhetorical_aggressiveness`	1–5	1 = procedural/hedged; 5 = hostile/geopolitical
`systemic_reach`	1–5	1 = product-specific; 5 = regime-challenging
`escalation_ultimatum`	1–5	1 = routine; 5 = retaliatory/rebalancing
`domestic_victimhood`	1–5	1 = no domestic pain; 5 = existential threat
`severity_score`	1–5	Mean of 4 dimensions (composite)
`reasoning`, `evidence`	—	LLM justification and supporting text
`n_parents_retrieved`	Integer	Parent chunks used for scoring

Table 23: Task B severity scores (raw, un-normalized). Anchors: DS3 = 1 (procedural); DS267 = 3 (assertive); DS574 = 5 (geopolitical). Post-hoc z-score normalization deferred until third-party scoring is complete.

Network Analysis Results

Year	Nodes	Edges	Conflict Density	Modularity	Comm.
1995	26	100	0.086	+0.196	3
2000	36	267	0.065	+0.019	3
2005	26	100	0.037	+0.182	4
2010	33	217	0.036	+0.031	2
2015	32	227	0.026	−0.052	3
2018	50	1,117	0.032	+0.086	4
2020	23	87	0.020	+0.173	2
2024	20	39	0.026	+0.200	1

Table 24: Selected annual network metrics from `wto_analysis_results_1995-2024.json`. Nodes = countries active in disputes that year; Comm. = communities (Louvain).

Document processing statistics (March 2026 — Final):

Metric	Value
Total PDFs processed	9,417 (63 pre-split PDFs manually combined)
Successfully classified	9,414 (99.97%)
Manual review required	3 files (non-English cross-references)
Document types	42 types
Date coverage	≈90% (multilingual + multi-part inheritance)
Third-party joinings	1,036 entries detected

Table 25: Final document processing statistics.

Dyadic Datasets

WTO Dispute Dyads

`wto_dyadic_v2.csv`

Attribute	Value
Unit of analysis	Case-level directed dyad
Rows	8,373
Columns	36
Unique cases	644

Table 26: `wto_dyadic_v2.csv` overview.

Each row is a directed country-pair relationship within a specific WTO case. `relationship` takes three values: `complainant-respondent` (684), `complainant-third_party` (3,946), `third_party-respondent` (3,743).

Key columns: `case` ("DS{N}"), `iso3_1`/`iso3_2`, `ccode1`/`ccode2`, `relationship`, plus 28 metadata columns inherited from `wto_cases_v2.csv`. See Section 3.5 for EU handling.

Bilateral Trade (WTO-filtered)

`bilateral_trade_wto.csv`

Attribute	Value
Unit of analysis	Directed dyad-year (exporter → importer)
Rows	545,333
Columns	31
Years	1995–2024

Table 27: `bilateral_trade_wto.csv` overview. WTO membership filter removes 337,310 non-member dyad-year rows. Uses dual EU representation: individual member states + `EUN` aggregate.

Trade Variables (from CEPII BACI HS92)

Gaulier, Guillaume, and Soledad Zignago. 2010. "BACI: International Trade Database at the Product-Level (the 1994-2007 Version)." *SSRN Electronic Journal* **Dataset:** https://www.cepii.fr/CEPII

Variable	Definition	Unit
`year`	Calendar year	Integer; 1995–2024
`exporter`	Exporter ISO3 code	String
`exporter_ccode`	Exporter COW numeric code	Integer
`importer`	Importer ISO3 code	String
`importer_ccode`	Importer COW numeric code	Integer
`total_trade_ij`	Total exports from i to j	Thousands USD
`n_products_ij`	Distinct HS6 products traded	Integer
`n_sections_ij`	Distinct HS sections traded	Integer; 1–21
`export_dependence`	`total_trade_ij / total_exports_i`	Ratio (0–1)
`trade_dependence`	$(\text{exports}_{ij} + \text{imports}_{ij})/\text{gdp}_i$	Ratio; unbounded
`total_exports_i`	Total exports of exporter i	Thousands USD

Table 28: Trade variables. Import-side columns omitted by design (redundant with the reverse row). `trade_dependence`: `NaN` for 2.4% lacking GDP data.

ATOP Alliance Variables

Leeds, Brett, et al. 2002. "Alliance Treaty Obligations and Provisions, 1815-1944." *International Interactions* 28 (3): 237–260 (2019–2024 forward-filled) `atopally`, `defense`, `offense`, `neutral`, `nonagg`, `consul`: all binary (0/1). Allied: 130,206 rows (23.9%). **Dataset:** http://www.atopdata.org

DESTA Trade Agreement Variables

Dür, Andreas, Leonardo Baccini, and Manfred Elsig. 2014. "The design of international trade agreements: Introducing a new dataset." *The Review of International Organizations* 9 (3): 353–375 **Dataset:** https://www.designoftradeagreements.org

Variable	Definition	Values / Notes
`label`	Trade agreement in force	0/1; has agreement: 37.6%
`number`	DESTA agreement ID	Integer; `NaN` if no agreement
`name`	Agreement name	String
`depth_index`	Additive depth (7 provisions)	Integer; 0–7
`depth_rasch`	Rasch-scaled depth	Continuous
`enforce01`	DSM strength index (std.)	Continuous; 0–6

Table 29: Key DESTA variables. Undirected by source; pipeline merges both sort orders. EUN rows all have `label=0` in this file (DESTA EUN data artifact; see Section 3.4).

Bilateral Trade by Section (WTO-filtered)

`bilateral_trade_section_wto.csv`

Attribute	Value
Unit of analysis	Directed dyad-year-section
Rows	6,374,486
Columns	14
Sections	21 HS sections
Missing values	None

Table 30: `bilateral_trade_section_wto.csv` overview.

Variable	Definition	Unit
`section_num`	HS section number (1–21)	Integer
`section_en`	HS section English name	String
`trade_value_ij_s`	Exports from i to j in section s	Thousands USD
`n_products_ij_s`	Distinct HS6 products in section s	Integer
`bilateral_sector_concentration`	`trade_value_ij_s / total_exports_i`	Ratio (0–1)
`sector_export_concentration`	`total_exports_i_s / total_exports_i`	Ratio (0–1)
`partner_sector_dependence`	`trade_value_ij_s / total_exports_i_s`	Ratio (0–1)

Table 31: Section-level trade variables. Used by the ERGM pipeline to compute disputed-sector trade values (t, t-1, t-2, t-3) for each dyadic case row.

DESTA Trade Agreement Panel

`desta_panel_1995_2025.csv`

Attribute	Value
Unit of analysis	Undirected dyad-year (alphabetically sorted)
Rows	235,011
Columns	21
Unique dyads	7,581
Years	1995–2025

Table 32: `desta_panel_1995_2025.csv` overview. Has agreement: 167,521 rows (71.3%); No agreement: 67,490 rows (28.7%).

Key variables: `country1`, `country2`, `year`, `label`, `iso3_1`, `iso3_2`, `baci_code1`, `baci_code2`, `ccode1`, `ccode2`, `number`, `base_treaty`, `name`, `entry_type`, `typememb`, `depth_index`, `depth_rasch`, `flexrigid`, `flexescape`, `enforce`, `enforce01`.

ISO mapping note: DESTA uses ISO 3166-1 numeric codes (USA=840) which differ from BACI codes (USA=842). Mapped via `country_converter` with `src='ISOnumeric'`. Taiwan (ISO=158) mapped to `TAW`.

ERGM Analysis Datasets

Four ERGM datasets built by `scripts/build_ergm_data.py`.

`wto_cases_enriched.csv` (644 rows, 44 cols)

Joins `wto_cases_v2.csv` with severity scores (`severity_scores_raw.csv`) and HS classification (`case_hs_sections.csv`). DS627–DS644 have `NaN` for all severity and HS columns.

Variable Group	Variables
Case metadata (from `wto_cases_v2.csv`)	All 29 original columns
Derived year	`consultation_year`: extracted from `consultations_requested`; falls back to `panel_requested`
Severity dimensions	`rhetorical_aggressiveness`, `systemic_reach`, `escalation_ultimatum`, `domestic_victimhood`, `severity_score`
HS classification	`hs_sections`, `title_hs_sections`, `case_type`, `extraction_method`, `confidence`
Policy flag	`is_horizontal_policy`: `True` if case spans all 21 HS sections

Table 33: Variable groups in `wto_cases_enriched.csv`.

`wto_dyadic_enriched.csv` (8,373 rows, 64 cols)

Joins `wto_dyadic_v2.csv` with case-level attributes, assigns `dyadic_severity` by relationship role, and adds disputed-sector bilateral trade for t, $t-1$, $t-2$, $t-3$.

Variable Group	Variables / Notes
Dyadic metadata	All 36 columns from `wto_dyadic_v2.csv`
Case attributes	`consultation_year`, `severity_score`, `hs_sections`, `case_type`, `is_horizontal_policy`, 4 severity dimensions
`dyadic_severity`	`complainant-respondent`: case `severity_score`; `third_party-respondent`: 0.5 (placeholder until TP scoring); `complainant-third_party`: 0
`most_severe_for_dyad_year`	Flags the most severe C-R case per (iso3_1, iso3_2, year)
Disputed sector trade	`disputed_trade_ij/ji_t0/t1/t2/t3` (Thousands USD) and `disputed_dep_ij/ji_t0/t1/t2/t3` (partner sector dependence) for the HS sections in each case, four time lags, both directions (16 cols)

Table 34: Variable groups in `wto_dyadic_enriched.csv`. Trade lags use backward imputation: 1995 → all NaN; 1996 → t-2 = t-3 = t-1; 1997 → t-3 = t-2.

`ergm_dyad_year_eun.csv` (545,344 rows, 190 cols)

The main ERGM panel. Universe = `bilateral_trade_wto` (all WTO-member directed dyad-years). EU treated as a unitary actor (`EUN`). Outcome: `has_dispute` = 1 when exporter filed a complaint against importer in year t (561 dispute dyad-years; 0.10%).

Variable Group	Variables / Notes
Base trade universe	All 31 columns from `bilateral_trade_wto.csv` (with EUN DESTA fix applied)
Dispute outcome	`has_dispute` (0/1), `dispute_count`, `case_id_max`, `severity_max`, `severity_avg`, `dyadic_severity_max`, `dyadic_severity_avg`
HS section dummies	`section_1` through `section_21` (from most severe case per dyad-year)
Disputed sector trade	`disputed_trade_ij/ji_t0/t1/t2/t3_max/avg` and `disputed_dep_ij/ji_t0/t1/t2/t3_max/avg` (32 cols)
Trade lags	`total_trade_ij`, `export_dependence`, `n_products_ij`, `n_sections_ij`, `trade_dependence` $\times$ 3 lags (t-1/t-2/t-3) = 15 cols. ATOP and DESTA kept contemporaneous only.
Node attributes (exporter, `_1`)	44 columns from `country_meta_1995_2024.csv` (29 excluded; see Section 1), contemporaneous year t, suffixed `_1`
Node attributes (importer, `_2`)	Same 44 columns, suffixed `_2`; also `n_complainant_t_2`, `n_respondent_t_2`, `n_tp_t_2`
Annual WTO counts	`n_complainant_t`, `n_respondent_t`, `n_tp_t`: annual (non-cumulative) dispute participation counts per country
TP covariate	`tp_with_j_comp_i`: has exporter historically acted as TP when importer was complainant? (binary, lagged 1 year); `tp_count_i_supporting_j`: cumulative count

Table 35: Variable groups in `ergm_dyad_year_eun.csv`. EUN node attributes: WTO participation columns (`complainant`, `respondent`, `cum_*`, `wto_member`, `wto_accession_year`) come directly from EUN's own `country_meta` rows (collective EU filings — do not aggregate from member states). GDP, pop, V-Dem, WGI etc. are filled from GDP-weighted member-state aggregation (`pop` summed; others population-weighted averaged). `eu_member` = 1, `euro_join_year` = 1999, `euro_member` = ($year \geq 1999$) hard-overridden at build step.

`ergm_dyad_year_eu_disagg.csv` (537,833 rows, 190 cols)

EU-disaggregated robustness version. All 7,511 EUN rows are removed. EU disputes expanded to individual member state rows using the three-category rule (Section 3.5). Has dispute: 3,471 rows (0.65%), reflecting EUN-only cases expanded to individual EU member states.

Same 190 columns as `ergm_dyad_year_eun.csv`. The higher dispute rate (0.64% vs. 0.10%) is by construction: EUN disputes originally counted as one row are now counted once per EU member state.

References

Bailey, Michael A., Anton Strezhnev, and Erik Voeten. 2017. "Estimating Dynamic State Preferences from United Nations Voting Data." *Journal of Conflict Resolution* 61 (2): 430–456.

Coppedge, Michael, John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, Jan Teorell, David Altman, Fabio Angiolillo, et al. 2025. *V-Dem Dataset v15.*

Dür, Andreas, Leonardo Baccini, and Manfred Elsig. 2014. "The design of international trade agreements: Introducing a new dataset." *The Review of International Organizations* 9 (3): 353–375.

Fjelstul, Joshua, Simon Hug, and Christopher Kilby. 2025. "Decision-making in the United Nations General Assembly: A comprehensive database of resolution-related decisions." *The Review of International Organizations.*

Gaulier, Guillaume, and Soledad Zignago. 2010. "BACI: International Trade Database at the Product-Level (the 1994-2007 Version)." *SSRN Electronic Journal.*

Kaufmann, Daniel, and Aart Kraay. 2025. "The Worldwide Governance Indicators : Methodology and 2024 Update." *SSRN Electronic Journal.*

Leeds, Brett, Jeffrey Ritter, Sara Mitchell, and Andrew Long. 2002. "Alliance Treaty Obligations and Provisions, 1815-1944." *International Interactions* 28 (3): 237–260.

Pemstein, Daniel, Kyle L. Marquardt, Eitan Tzelgov, Yi-ting Wang, Juraj Medzihorsky, Joshua Krusell, Farhad Miri, and Johannes von Römer. 2025. *The V-Dem Measurement Model: Latent Variable Analysis for Cross-National and Cross-Temporal Expert-Coded Data.* Technical report V-Dem Working Paper No. 21. University of Gothenburg: Varieties of Democracy Institute.

Singer, J. David. 1988. "Reconstructing the correlates of war dataset on material capabilities of states, 1816–1985." _Eprint: https://doi.org/10.1080/03050628808434695, *International Interactions* 14 (2): 115–132.

Singer, J. David, Stuart A. Bremer, and John Stuckey. 2012. "Capability distribution, uncertainty, and major power war, 1820–1965 1." In *Advancing Peace Research,* 161–174. Routledge.